Final Assignment : Logistic Regression and Weighted Logistic Regression

Basic information about the dataset

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
from sklearn.model_selection import train_test_split,cross_val_score
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report,precision_score, recall_score,
f1_score,precision_recall_curve
```

```
ccf = pd.read_csv('/content/credit_card_fraud_10k.csv')
```

```
print(ccf.head())
```

```
   transaction_id  amount  transaction_hour merchant_category  \
0               1   84.47                22       Electronics
1               2  541.82                 3            Travel
2               3  237.01                17           Grocery
3               4  164.33                 4           Grocery
4               5   30.53                15              Food

   foreign_transaction  location_mismatch  device_trust_score  \
0                    0                  0                  66
1                    1                  0                  87
2                    0                  0                  49
3                    0                  1                  72
4                    0                  0                  79

   velocity_last_24h  cardholder_age  is_fraud
0                  3              40         0
1                  1              64         0
2                  1              61         0
3                  3              34         0
4                  0              44         0
```

```
ccf.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 10 columns):
 #   Column               Non-Null Count  Dtype
```

```
---   ------                --------------  -----
 0   transaction_id       10000 non-null  int64
 1   amount               10000 non-null  float64
 2   transaction_hour     10000 non-null  int64
 3   merchant_category    10000 non-null  object
 4   foreign_transaction  10000 non-null  int64
 5   location_mismatch    10000 non-null  int64
 6   device_trust_score   10000 non-null  int64
 7   velocity_last_24h    10000 non-null  int64
 8   cardholder_age       10000 non-null  int64
 9   is_fraud             10000 non-null  int64
dtypes: float64(1), int64(8), object(1)
memory usage: 781.4+ KB
```

```
print(ccf.describe())
```

```
       transaction_id        amount  transaction_hour
foreign_transaction  \
count     10000.00000  10000.000000      10000.000000
10000.000000
mean       5000.50000    175.949849         11.593300
0.097800
std        2886.89568    175.392827          6.922708
0.297059
min           1.00000      0.000000          0.000000
0.000000
25%        2500.75000     50.905000          6.000000
0.000000
50%        5000.50000    122.095000         12.000000
0.000000
75%        7500.25000    242.480000         18.000000
0.000000
max       10000.00000   1471.040000         23.000000
1.000000

       location_mismatch  device_trust_score  velocity_last_24h  \
count       10000.000000        10000.000000       10000.000000
mean            0.085700           61.798900           2.008900
std             0.279935           21.487053           1.432559
min             0.000000           25.000000           0.000000
25%             0.000000           43.000000           1.000000
50%             0.000000           62.000000           2.000000
75%             0.000000           80.000000           3.000000
max             1.000000           99.000000           9.000000

       cardholder_age      is_fraud
count    10000.000000  10000.000000
mean        43.468700      0.015100
std         14.979147      0.121957
min         18.000000      0.000000
```

```
25%          30.000000      0.000000
50%          44.000000      0.000000
75%          56.000000      0.000000
max          69.000000      1.000000
```

```
ccf.drop(columns=['transaction_id'], inplace=True)
```

```
ccf.shape
```

```
(10000, 9)
```

Data Cleaning

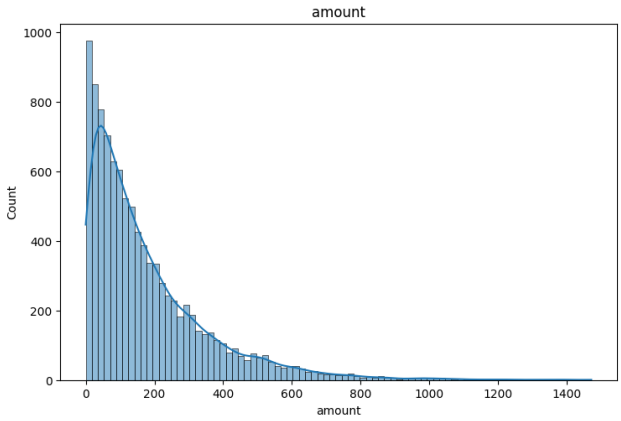
```
ccf_copy = ccf.copy()

print(f'Number of dublicate rows are: {ccf_copy.duplicated().sum()}')
```

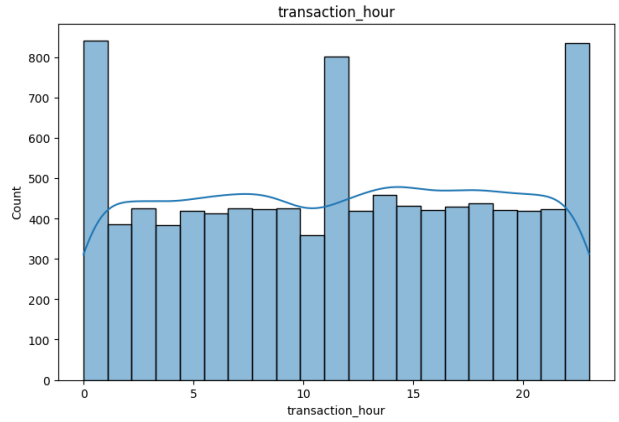
```
Number of dublicate rows are: 0
```

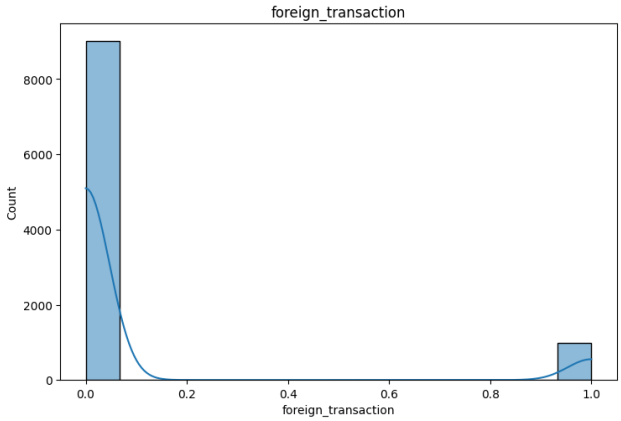
```
print(f'Number of null values in the dataset:\
n{ccf_copy.isnull().sum()}')
```

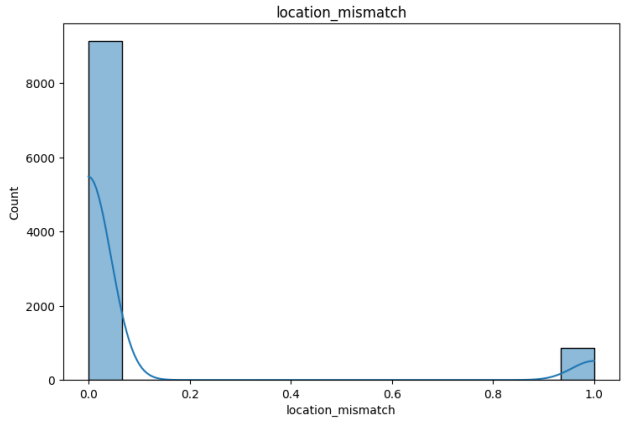
```
Number of null values in the dataset:
amount                  0
transaction_hour        0
merchant_category       0
foreign_transaction     0
location_mismatch       0
device_trust_score      0
velocity_last_24h       0
cardholder_age          0
is_fraud                0
dtype: int64
```

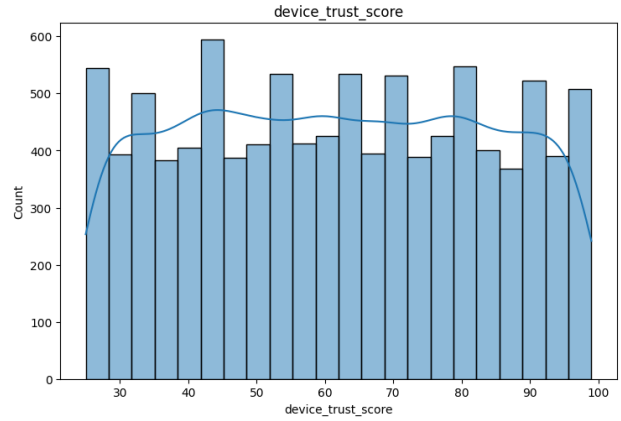
```
num_cols = ccf_copy.select_dtypes(include =
['int64','float64']).columns
num_cols = num_cols.drop('is_fraud')

plt.figure(figsize=(15, 25))

for i, col in enumerate(num_cols, start=1):
    plt.subplot(5, 2, i)
    sns.histplot(ccf_copy[col], kde=True)
    plt.title(col)

plt.tight_layout()
plt.show()
```

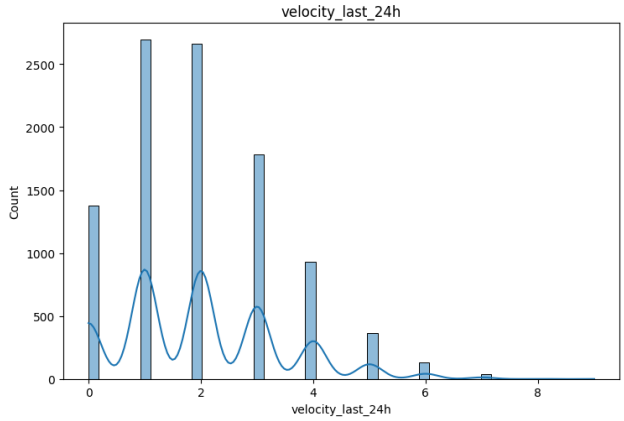

amount
Count
amount

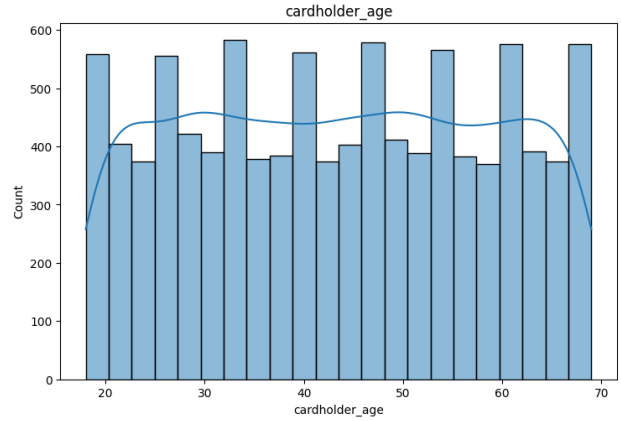

transaction_hour
Count
transaction_hour


foreign_transaction
Count
foreign_transaction

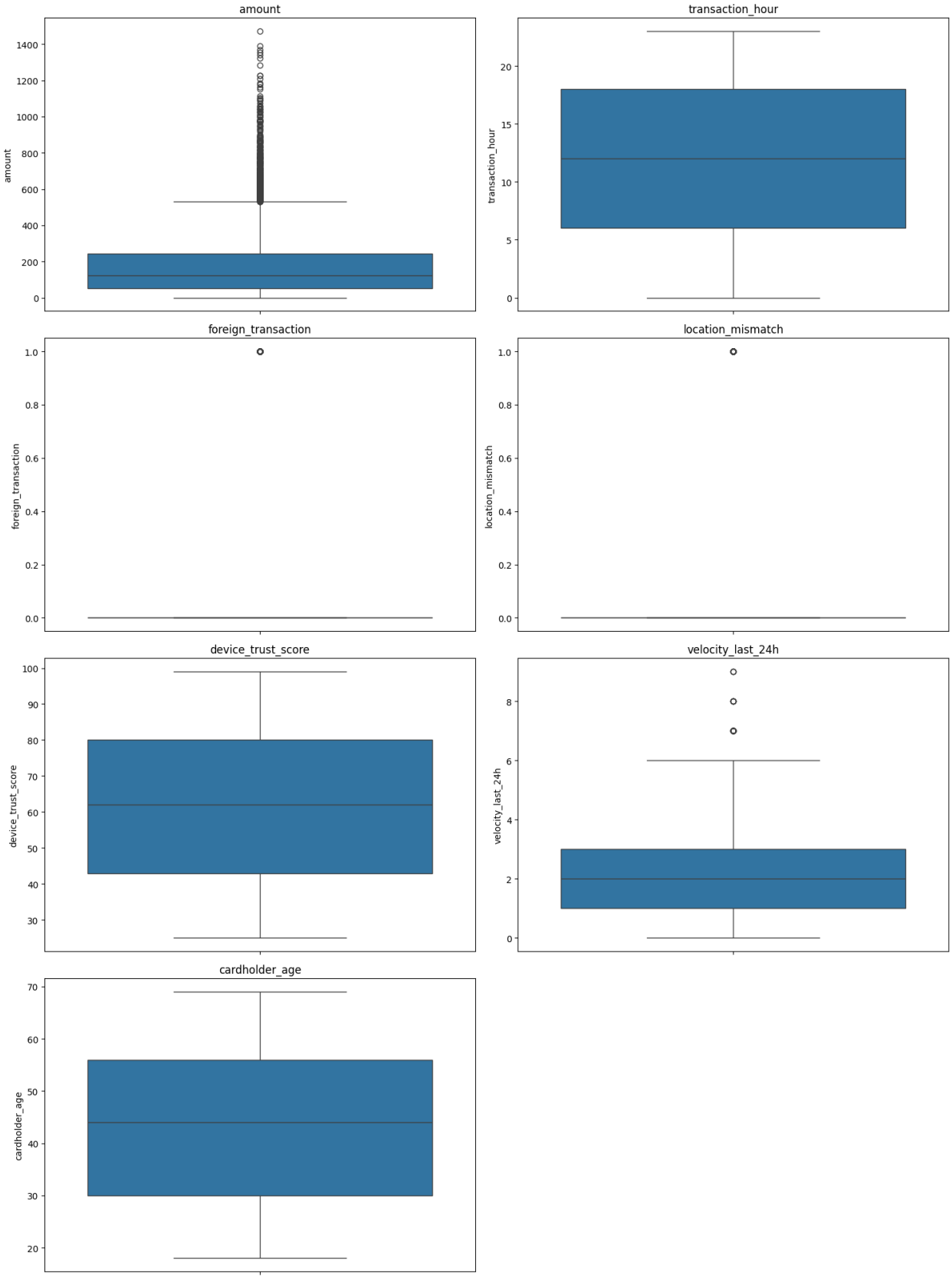

location_mismatch
Count
location_mismatch


device_trust_score
Count
device_trust_score


velocity_last_24h
Count
velocity_last_24h


cardholder_age
Count
cardholder_age

```
plt.figure(figsize=(15, 25))

for i, col in enumerate(num_cols, start=1):
    plt.subplot(5, 2, i)
    sns.boxplot(ccf_copy[col])
    plt.title(col)

plt.tight_layout()
plt.show()
```


amount
amount
0
200
400
600
800
1000
1200
1400
transaction_hour
transaction_hour
0
5
10
15
20
foreign_transaction
foreign_transaction
0.0
0.2
0.4
0.6
0.8
1.0
location_mismatch
location_mismatch
0.0
0.2
0.4
0.6
0.8
1.0
device_trust_score
device_trust_score
30
40
50
60
70
80
90
100
velocity_last_24h
velocity_last_24h
0
2
4
6
8
cardholder_age
cardholder_age
20
30
40
50
60
70

```
ccf_copy[num_cols].skew()
```

```
amount                  1.919265
transaction_hour       -0.026581
foreign_transaction     2.708422
location_mismatch       2.960570
device_trust_score      0.011191
velocity_last_24h       0.707884
cardholder_age          0.003868
dtype: float64
```

```
ccf_copy['is_fraud'].value_counts() * 100
```

```
is_fraud
0    984900
1     15100
Name: count, dtype: int64
```

Data Preprocessing

```
# Separating dataset into X and y
X = ccf_copy.drop(columns=['is_fraud'])  # example target
y = ccf_copy['is_fraud']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# Log transformation to reduce skewness
X_train['amount'] = np.log1p(X_train['amount'])
X_test['amount']  = np.log1p(X_test['amount'])

X_train[num_cols].skew()
```

```
amount                 -0.787350
transaction_hour       -0.027348
foreign_transaction     2.695231
location_mismatch       2.976731
device_trust_score     -0.000353
velocity_last_24h       0.708300
cardholder_age          0.002899
dtype: float64
```

```
num_cols = num_cols.drop(['foreign_transaction','location_mismatch'])

scaler = StandardScaler()

X_train[num_cols] = scaler.fit_transform(X_train[num_cols])
X_test[num_cols]  = scaler.transform(X_test[num_cols])
```

```
encoder = OneHotEncoder(drop='first', sparse_output=False)

X_train_cat = pd.DataFrame(
    encoder.fit_transform(X_train[['merchant_category']]),
    columns=encoder.get_feature_names_out(['merchant_category']),
    index=X_train.index
)

X_test_cat = pd.DataFrame(
    encoder.transform(X_test[['merchant_category']]),
    columns=encoder.get_feature_names_out(['merchant_category']),
    index=X_test.index
)

# Concatenate with numerical columns
X_train_final = pd.concat([X_train[:], X_train_cat], axis=1)
X_test_final  = pd.concat([X_test[:], X_test_cat], axis=1)

X_train_final.drop(columns=['merchant_category'], inplace=True)
X_test_final.drop(columns=['merchant_category'], inplace=True)
```

Model Training and Evaluation Matrices

Standard Logistic Regression

```
log_reg = LogisticRegression(max_iter=1000)
log_reg.fit(X_train_final, y_train)

y_pred = log_reg.predict(X_test_final)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall   :",recall_score(y_test, y_pred))
print("F1 Score :",f1_score(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test,
y_pred))
```

```
Accuracy: 0.993
Precision: 0.9444444444444444
Recall   : 0.5666666666666667
F1 Score : 0.7083333333333334

Confusion Matrix:
 [[1969    1]
 [  13   17]]

Classification Report:
               precision    recall  f1-score   support
```

```
           0       0.99      1.00      1.00      1970
           1       0.94      0.57      0.71        30

    accuracy                           0.99      2000
   macro avg       0.97      0.78      0.85      2000
weighted avg       0.99      0.99      0.99      2000
```

Weighted Logistic Regression

```
n0 = np.sum(y_train == 0)
n1 = np.sum(y_train == 1)

w0 = len(y_train) / (2 * n0)
w1 = len(y_train) / (2 * n1)

print("Class weights:", w0, w1)
```

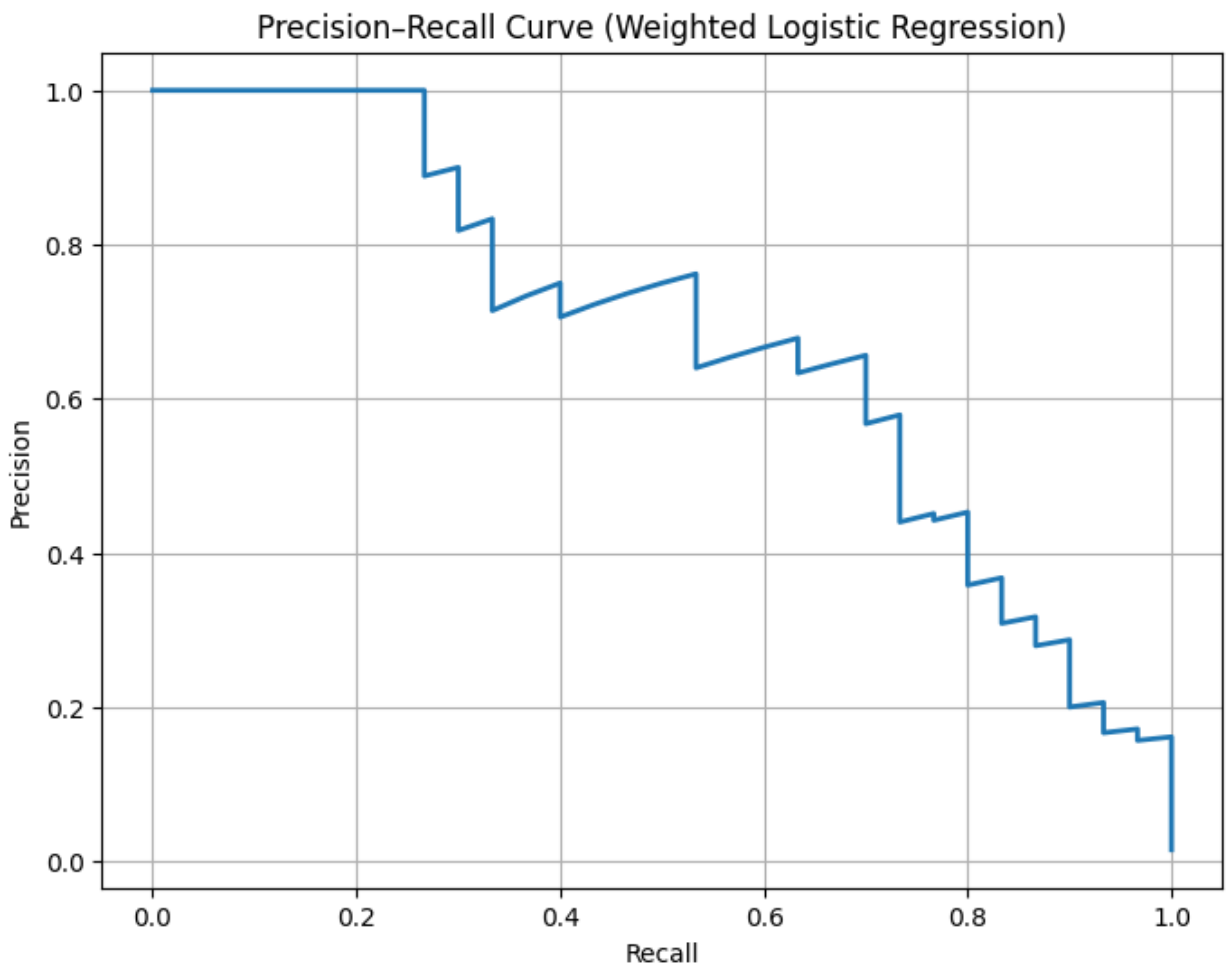
```
Class weights: 0.5076786394212464 33.05785123966942
```

```
def sigmoid(z):
    return 1 / (1 + np.exp(-np.clip(z, -500, 500)))
def compute_gradients(X, y, y_hat,w0,w1):
    m = len(y)
    weights = np.where(y == 1, w1, w0)
    dw = (1 / m) * np.dot(X.T, weights * (y_hat - y))
    db = (1 / m) * np.sum(weights * (y_hat - y))
    return dw, db
def train_weighted_logistic_regression(X, y, lr=0.005,
epochs=6000,w0=w0,w1=w1):
    m, n = X.shape
    w = np.zeros(n)
    b = 0
    for _ in range(epochs):
        z = np.dot(X, w) + b
        y_hat = sigmoid(z)
        dw, db = compute_gradients(X, y, y_hat,w0,w1)
        w -= lr * dw
        b -= lr * db
    return w, b

w_weighted, b_weighted = train_weighted_logistic_regression(
    X_train_final,
    y_train
)

y_proba = sigmoid(np.dot(X_test_final, w_weighted) + b_weighted)
y_pred  = (y_proba >= 0.5).astype(int)

print("\nManual Weighted Logistic Regression\n")
```

```
print("Accuracy :", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall   :", recall_score(y_test, y_pred))
print("F1 Score :", f1_score(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

```
Manual Weighted Logistic Regression

Accuracy : 0.863
Precision: 0.09868421052631579
Recall   : 1.0
F1 Score : 0.17964071856287425

Confusion Matrix:
[[1696  274]
 [   0   30]]
```

```
precision, recall, thresholds = precision_recall_curve(y_test,
y_proba)

plt.figure(figsize=(8, 6))
plt.plot(reall := recall, precision, linewidth=2)
plt.xlabel("Recall")
plt.ylabel("Precision")
plt.title("Precision–Recall Curve (Weighted Logistic Regression)")
plt.grid()
plt.show()
```




```
# Find thresholds where recall >= 0.80
target_recall = 0.80

valid_idx = np.where(recall >= target_recall)[0]

# precision_recall_curve returns one extra recall value
# thresholds has length = len(recall) - 1
valid_idx = valid_idx[valid_idx < len(thresholds)]

best_idx = valid_idx[-1]   # highest threshold satisfying recall ≥
0.80
best_threshold = thresholds[best_idx]

print("Threshold for at least 80% recall:", best_threshold)
print("Precision at this threshold:", precision[best_idx])
print("Recall at this threshold:", recall[best_idx])

Threshold for at least 80% recall: 0.8152703176285074
Precision at this threshold: 0.4528301886792453
Recall at this threshold: 0.8
```

```python
y_pred_80_recall = (y_proba >= best_threshold).astype(int)

print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_80_recall))

print("\nFraud Precision:", precision_score(y_test, y_pred_80_recall))
print("Fraud Recall   :", recall_score(y_test, y_pred_80_recall))
print("Fraud F1       :", f1_score(y_test, y_pred_80_recall))
```

```
Confusion Matrix:
[[1941   29]
 [   6   24]]

Fraud Precision: 0.4528301886792453
Fraud Recall   : 0.8
Fraud F1       : 0.5783132530120482
```